

Generative AI Applications for Managers

Estimated time: 30 minutes

Welcome to the Generative AI Applications for Managers.

Learning objectives

After completing this lab, you will be able to:

- Explore how generative AI can be used to get insights and recommendations for leaders and managers.
- Demonstrate examples of using generative AI for creating dashboards.

Introduction

Leaders and managers are increasingly leveraging generative AI to enhance decision-making processes and streamline operations within their organizations. By harnessing the power of advanced language models, leaders can utilize natural language understanding and generation to automate routine tasks, analyze vast amounts of data for actionable insights, and generate human-like text for various applications. Generative AI supports leaders in formulating creative solutions, improving communication, and fostering innovation by providing valuable assistance.

In this lab, you will work with ChatGPT and Google Spreadsheet to demonstrate the use of generative AI by managers.

If you have access to Microsoft Excel, you can consider working in an Excel Workbook, instead of Google Spreadsheet.

Accessing Google Spreadsheets

To work with Google Spreadsheets, you need to have a Google account. If you do not have a Google account, then visit the following page and follow the steps to create an account.

<https://support.google.com/accounts/answer/27441?hl=en>

Log in to ChatGPT

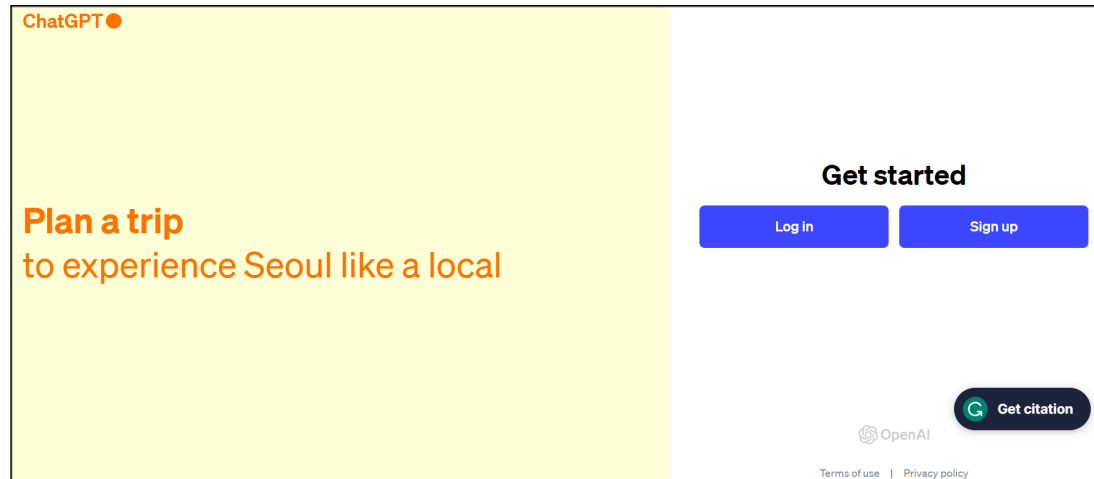
1. Launch [ChatGPT](#) and sign in.

Note: If you cannot access ChatGPT by clicking on the link, copy the following URL and paste it into a browser.

<https://chat.openai.com/>

2. Once launched, you must sign up or login.

Note: You can create a free account on ChatGPT and use GPT-3.5 features for free.



- a. To sign up, you will be required to enter your email ID and password. You will receive an automated verification email. After you click verification, you will need to enter your personal details and follow the steps to complete the sign up on your first login.
- b. After sign up, you will reach the login page, as shown below.

Welcome back

Email address

Continue

Don't have an account? [Sign up](#)

OR



Continue with Google

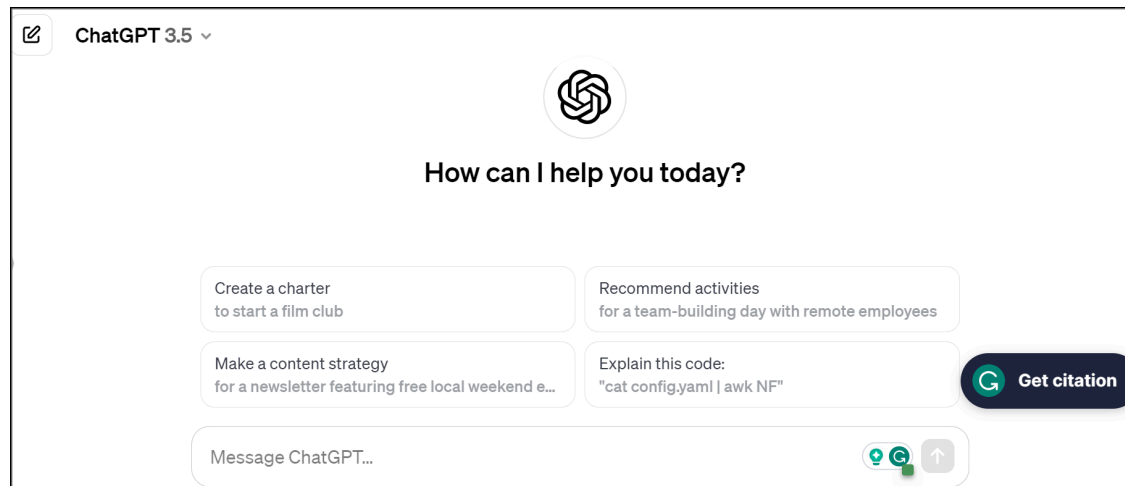


Continue with Microsoft Account



Continue with Apple

3. After login, you will view the OpenAI chatbot platform, ChatGPT, as shown below.



Scenario

In this lab, you'll work with an example to create dashboards for analysis of (fictitious) data for car sales. The car sales data is provided in the sheet - [SalesData](#).

Exercise 1: Generate a sales performance dashboard

In this exercise, we will analyze monthly sales performance and engage in a conversation with ChatGPT to get insights and recommendations.

Step 1: Open the SalesData sheet in a Google Spreadsheet

1. Right-click on the link [SalesData](#) and open it in a new tab. This will download the file to your system.



We've opened your file for quick and easy viewing right in Microsoft



SalesData - View-only

File

Home

Insert

Share

Page Layout

Formulas

Data



12



B



16



fx

Passenger

	A	B	C	D
1	Manufacturer	Model	Sales_in_thousands	__year_resale_v
2	Acura	Integra	16.919	
3	Acura	TL	39.384	
4	Acura	CL	14.114	
5	Acura	RL	8.588	

6	Audi	A4	20.397
7	Audi	A6	18.78
8	Audi	A8	1.38
9	BMW	323i	19.747
10	BMW	328i	9.231
11	BMW	528i	17.527
12	Buick	Century	91.561
13	Buick	Regal	39.35
14	Buick	Park Avenue	27.851
15	Buick	LeSabre	83.257
16	Cadillac	DeVille	63.729
17	Cadillac	Seville	15.943
18	Cadillac	Eldorado	6.536



< > ≡ Car Sales data +

Workbook Statistics

2. To open Google Spreadsheets, navigate to the [Google Sheets](#) webpage. Sign in with your Google account. Enter your **email ID** or **phone number** and click on **Next**. Then enter your password and again click on **Next**.

The image displays two sequential steps of the Google Sheets sign-in process.

Step 1: Sign in screen

- Header: Google logo, "Sign in to continue to Sheets".
- Input field: "Email or phone" with the placeholder text "XXXXXXXXXX@gmail.com". This field is highlighted with a red box.
- Link: "Forgot email?".
- Text: "Not your computer? Use Guest mode to sign in privately. [Learn more about using Guest mode](#)".
- Buttons: "Create account" and a blue "Next" button. The "Next" button is highlighted with a red box.

Step 2: Welcome screen

- Header: Google logo, "Welcome".
- User selection: A dropdown menu showing "xxxxxxxxxx@gmail.com" with a downward arrow. This area is highlighted with a red box.
- Input field: "Enter your password". This field is highlighted with a red box.
- Checkbox: "Show password" with an unchecked checkbox.
- Link: "Forgot password?".
- Button: A blue "Next" button. This button is highlighted with a red box.

3. Once you have logged in with your Google account, click on **Google apps** and then click on **Sheets**.



Relaun

Gmail

Images



Finance



Docs



Shee



Adobe



Add shortcut



Slides



Books



Blog



Keep



Jamboard



Ear



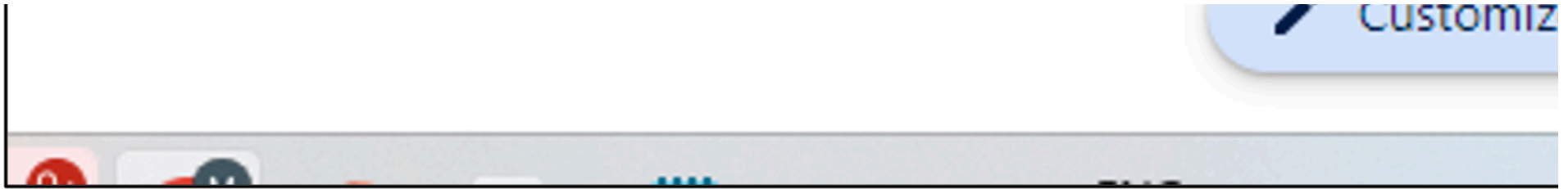
Saved



Arts and C...



Google



4. Click on **Blank spreadsheet**.



docs.google.com/spreadsheets/u/0/



Sheets



Start a new spreadsheet



Blank spreadsheet

To do

	Task	Due
☐	Change the website layout	10/10/2020
☐	Test new user interface	10/10/2020
☐		
☐		
☐		
☐		
☐		
☐		
☐		

To-do list

Earlier



Table of Content (TOC)_Mod



SOCH LMS Content Develop



Final Details of Videos.xlsx

5. Click on **File > Open**.



docs.google.com/spreadsheets/d/1EZTyY



Untitled spreadsheet



File

Edit

View

Insert

Format

Data

Tools

Ex



New



Open

Ctrl+O



Import



Make a copy



Share



Email

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6	 Download	
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9	 Rename	
10	 Move to trash	
11		

6. Next, click on **Upload**. Click **Browse** to navigate to the Car Sales Data sheet on your computer.



Open a file

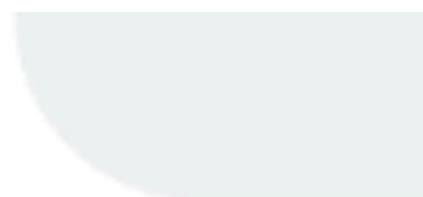
Recent

My Drive

Shared with me

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or dr

docs.google.com/spreadsheets/d/1OT_NnVDA59P9ZqZbsc9apolB0FvMxRst_5457xk1mo/edit#gid=381420575

Car Sales Data

File Edit View Insert Format Data Tools Extensions Help

100% 123 Default... 10 B I A

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Manufacturer	Model	Sales_in_thousands	_year_resale_value	Vehicle_type	Price_in_thousands	Engine_size	Horsepower	Wheelbase	Width	Length	Curb_weight	Fuel_capacity
2	Acura	Integra	16.919	16.36	Passenger	21.5	1.8	140	101.2	67.3	172.4	2.639	13.2
3	Acura	TL	39.384	19.875	Passenger	28.4	3.2	225	108.1	70.3	192.9	3.517	17.2
4	Acura	CL	14.114	18.225	Passenger		3.2	225	106.9	70.6	192	3.47	17.2
5	Acura	RL	8.588	29.725	Passenger	42	3.5	210	114.6	71.4	196.6	3.85	18
6	Audi	A4	20.397	22.255	Passenger	23.99	1.8	150	102.6	68.2	178	2.998	16.4
7	Audi	A6	18.78	23.555	Passenger	33.95	2.8	200	108.7	76.1	192	3.561	18.5
8	Audi	A8	1.38	39	Passenger	62	4.2	310	113	74	198.2	3.902	23.7
9	BMW	323i	19.747		Passenger	26.99	2.5	170	107.3	68.4	176	3.179	16.6
10	BMW	328i	9.231	28.675	Passenger	33.4	2.8	193	107.3	68.5	176	3.197	16.6
11	BMW	528i	17.527	36.125	Passenger	38.9	2.8	193	111.4	70.9	188	3.472	18.5
12	Buick	Century	91.561	12.475	Passenger	21.975	3.1	175	109	72.7	194.6	3.368	17.5
13	Buick	Regal	39.35	13.74	Passenger	25.3	3.8	240	109	72.7	196.2	3.543	17.5
14	Buick	Park Ave	27.851	20.19	Passenger	31.965	3.8	205	113.8	74.7	206.8	3.778	18.5
15	Buick	LeSabre	83.257	13.36	Passenger	27.885	3.8	205	112.2	73.5	200	3.591	17.5
16	Cadillac	DeVille	63.729	22.525	Passenger	39.895	4.6	275	115.3	74.5	207.2	3.978	18.5
17	Cadillac	Seville	15.943	27.1	Passenger	44.475	4.6	275	112.2	75	201		18.5
18	Cadillac	Eldorado	6.536	25.725	Passenger	39.665	4.6	275	108	75.5	200.6	3.843	19
19	Cadillac	Catera	11.185	18.225	Passenger	31.01	3	200	107.4	70.3	194.8	3.77	18
20	Cadillac	Escalade	14.785		Car	46.225	5.7	255	117.5	77	201.2	5.572	30
21	Chevrolet	Cavalier	145.519	9.25	Passenger	13.26	2.2	115	104.1	67.9	180.9	2.676	14.3
22	Chevrolet	Malibu	135.126	11.225	Passenger	16.535	3.1	170	107	69.4	190.4	3.051	15

Car Sales data

Step 2: Identify generic approach for creating a dashboard

2. Let's consider prompting ChatGPT to create a dashboard for the car sales data. Enter the following prompt.

I have some car sales data in Google spreadsheet with the following columns - Manufacturer, Model, Sales_in_thousands, _year_resale_value, Vehicle_type, Price_in_thousands, Engine_size, Horsepower,

3. After you enter the above prompt, ChatGPT will ask for more details for better understanding.

ChatGPT 3.5 ▾

MA

You

I have some car sales data
Manufacturer, Model, Sales
Price_in_thousands, Engine
Curb_weight, Fuel_capacity



ChatGPT

Great! It looks like you have
Spreadsheet. How can I as

analysis, or any particular t
in mind, and I'll do my best



Message ChatGPT...

4. You can further prompt ChatGPT with the following statement.

I want to create a sales dashboard for a spreadsheet from the car sales data provided. Can you help me?

5. Click the **Send Message** icon to generate the output. The output produced at our end is shown here. In the output, ChatGPT has provided a stepwise approach to creating a dashboard.



You

I want to create a sales dashboard for a spreadsheet from the car sales data provided. Can you help me?



ChatGPT

Certainly! Creating a sales dashboard can be a great way to visualize and analyze your car sales data effectively. Here are some steps you can follow using Google Sheets:

Step 1: Data Preparation

Make sure your data is organized properly. Ensure there are no empty cells, and each column has appropriate headers.

Step 2: Choose Key Metrics

Identify the key metrics you want to highlight in your sales dashboard. Common metrics for car sales include total sales, average price, top-selling models, etc.

Step 3: Create Pivot Tables

Use pivot tables to summarize and analyze your data. You can create pivot tables for total sales, average price, or any other metrics you've identified.

1. Insert Pivot Table:

- Highlight your data.
- Go to **Data > Pivot table**.
- Choose where to place the pivot table.

2. Configure Pivot Table:

- Drag the fields you need (e.g., Month, Manufacturer, Model) to Rows and Values.
- Choose the aggregation method for the values (e.g., sum for total sales).

Step 4: Create Charts

Visualize your data with charts. Google Sheets offers various chart types such as bar charts, line charts, and pie charts.

1. Insert Chart:

- Select the data you want to include in the chart.
- Go to **Insert > Chart**.



2. Configure Chart:

- Choose the chart type.
- Adjust the chart settings, titles, and legends.

Step 5: Combine Charts into a Dashboard

You can create a separate sheet for your dashboard and arrange your charts to provide a comprehensive view of your data.

1. Insert New Sheet:

- Click on the **+** button at the bottom to add a new sheet.

2. Copy and Paste Charts:

- Copy the charts from the individual sheets and paste them into your dashboard sheet.

3. Organize the Dashboard:

- Arrange the charts in a way that makes sense for your analysis.

Step 3: Create Pivot Tables

Considering the stepwise approach suggested by ChatGPT, we first need to create the pivot tables to summarize and analyze the data.

1. Pivot tables allow you to summarize and analyze your data quickly. Let's create a pivot table to gather insights. Open the spreadsheet named **"Car sales data"**. Select the entire data, click on **Insert**, and then click on **Pivot table**.



Car sales data



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Audi

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9

BMW

323i

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BMW

323i



Cells



Rows



Columns



Sheet

Shift+F11



Chart



Pivot table



Image



Drawing



Function

10	BMW	328i
11	BMW	528i
12	Buick	Centurion
13	Buick	Regal
14	Buick	Park Avenue
15	Buick	LeSabre
16	Cadillac	DeVille
17	Cadillac	Seville
18	Cadillac	Eldorado

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☒ Checkbox

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+ ≡ Sheet1 ▼ Sheet2 ▼ Pivot Table 1 ▼

2. Click on **Create** to start creating the pivot table.



Car sales data



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Manufacturer

	A	B	C	D	
1	Manufacturer	Model	Sales_in_thousa	__year_resale_v	Vehi
2	Acura	Integra	16.919	16.36	Pass
3	Acura	TL	39.384	19.875	Pass
4	Acura	CL	14.114	18.225	Pass
5	Acura	RL	8.588	29.725	Pass
6	Audi	A4	20.397	22.255	Pass
7	Audi	A6	18.78	23.555	Pass
8	Audi	A8	1.38	39	Pass
9	BMW	323i	19.747		Pass

10	BMW	328i	9.231	28.675	Pass
11	BMW	528i	17.527	36.125	Pass
12	Buick	Century	91.561	12.475	Pass
13	Buick	Regal	39.35	13.74	Pass
14	Buick	Park Avenue	27.851	20.19	Pass
15	Buick	LeSabre	83.257	13.36	Pass
16	Cadillac	DeVille	63.729	22.525	Pass
17	Cadillac	Seville	15.943	27.1	Pass
18	Cadillac	Eldorado	6.536	25.725	Pass

3. Drag **Manufacturer** to **Rows** and **Sales_in_thousands** to the **Values** field.



Car sales data



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4. Click on the arrow under the **Values** field and select **SUM**.



Car sales data



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Default

A1

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Manufacturer

	A	B	C	D	E
1	Manufacturer	SUM of Sales_in			
2	Acura	79.005			
3	Audi	40.557			
4	BMW	46.505			
5	Buick	242.019			
6	Cadillac	112.178			
7	Chevrolet	554.365			
8	Chrysler	201.721			
9	Dodge	910.149			
10	Ford	2022.635			

11	Honda	592.674
12	Hyundai	137.326
13	Infiniti	23.713
14	Jaguar	15.467
15	Jeep	293.153
16	Lexus	106.843
17	Lincoln	85.634
18	Mercedes-B	117.125
19	Mercury	237.999
20	 ubishi	180.895

5. The pivot table is ready. You can also change the name of this table.

Car sales data

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A1 fx Manufacturer

	A	B	C	D
1	Manufacturer	SUM of Sales in		
2	Acura	79.005		
3	Audi	40.557		
4	BMW	46.505		
5	Buick	242.019		
6	Cadillac	112.178		
7	Chevrolet	554.365		
8	Chrysler	201.721		
9	Dodge	910.149		
10	Ford	2022.635		
11	Honda	592.674		
12	Hyundai	137.326		
13	Infiniti	23.713		
14	Jaguar	15.467		
15	Jeep	293.153		
16	Lexus	106.843		
17	Lincoln	85.634		
18	Mercedes-B	117.125		
19	Mercury	237.999		
20	Subaru	180.895		

Sheet1 Sheet2 Pivot Table 1

6. To create another pivot table, go back to the data. Select the entire data, click on **Insert**, and then click on **Pivot table**.

Car sales data

File Edit View **Insert** Format Data Tools Extensions Help

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Manufacturer Model Year Price In thousands Engine size Horsepower Wheelbase Width Length Curb weight Fuel economy city

1 Acura Integra 2001 21.5 1.8 140 101.2 67.3 172.4 2.639

2 Acura TL 2001 28.4 3.2 225 108.1 70.3 192.9 3.517

3 Acura CL 2001 32 225 106.9 70.6 192 3.47

4 Acura RL 2001 42 3.5 210 114.6 71.4 196.6 3.85

5 Audi A4 2001 23.99 1.8 150 102.6 68.2 178 2.998

6 Audi A6 2001 33.95 2.8 200 108.7 76.1 192 3.561

7 Audi A8 2001 62 4.2 310 113 74 198.2 3.902

8 BMW 323i 2001 26.99 2.5 170 107.3 68.4 176 3.179

9 BMW 328i 2001 33.4 2.8 193 107.3 68.5 176 3.197

10 BMW 528i 2001 38.9 2.8 193 111.4 70.9 188 3.472

11 Buick Century 2001 21.975 3.1 175 109 72.7 194.6 3.368

12 Buick Regal 2001 25.3 3.8 240 109 72.7 196.2 3.543

13 Buick Park Avenue 2001 31.965 3.8 205 113.8 74.7 206.8 3.778

14 Buick LeSabre 2001 27.885 3.8 205 112.2 73.5 200 3.591

15 Cadillac Deville 2001 39.895 4.6 275 115.3 74.5 207.2 3.978

16 Cadillac Seville 2001 44.475 4.6 275 112.2 75 201 3.843

17 Cadillac Eldorado 2001 39.665 4.6 275 108 75.5 200.6 3.843

Sheet1 Sheet2 Pivot Table 1

Count: 2,477

7. This time, drag **Vehicle_type** to **Rows** and **Price_in_thousands** to the **Values** field.



Car sales data



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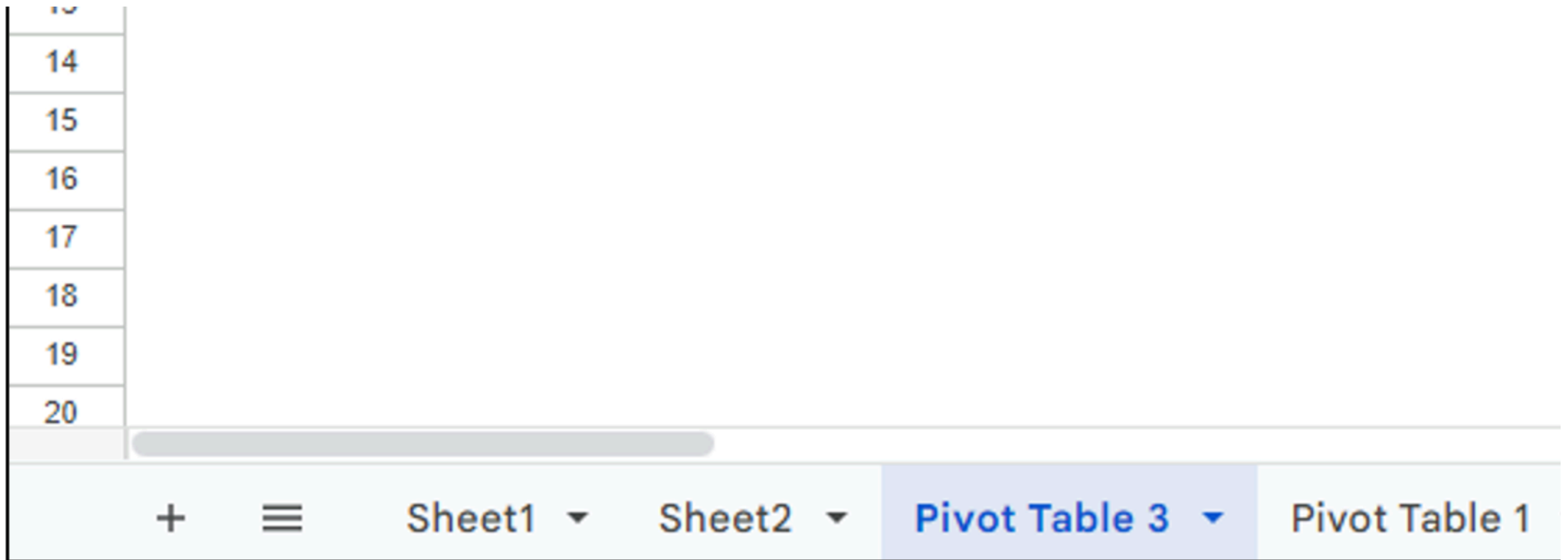
A1



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Vehicle_type

	A	B	C	D	E
1	Vehicle_type	COUNTA of Pric			
2	Car	40			
3	Passenger	115			
4	Grand Total	155			
5					
6					
7					
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9					
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12					
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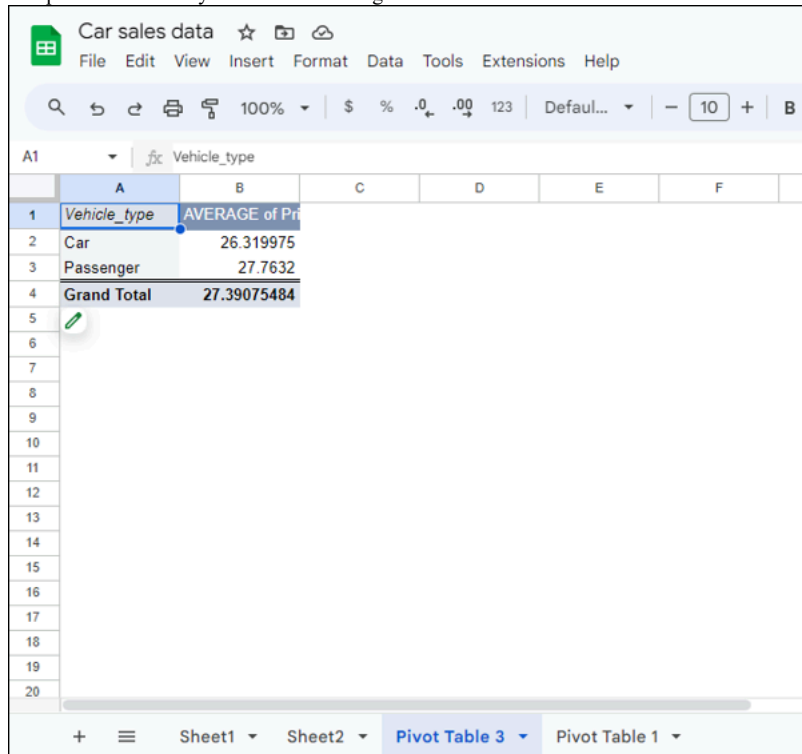
8. Click on the arrow under the **Values** field and select **Average**.

This image shows a Google Sheets interface with a Pivot Table. The Pivot Table is located in the range A1:B4 and has the following data:

Vehicle_type	COUNTA of Price
Car	40
Passenger	115
Grand Total	155

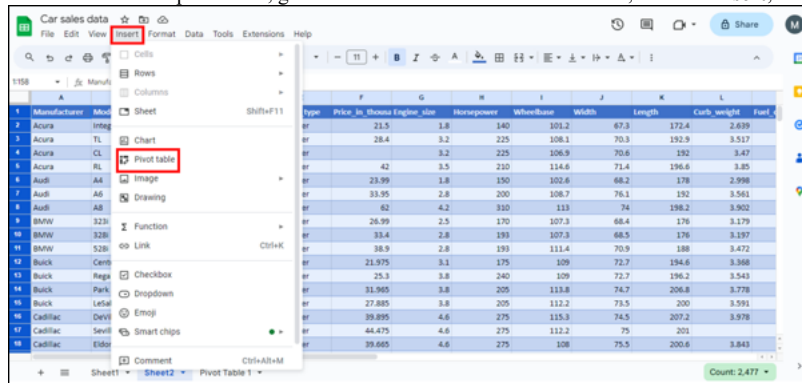
The 'Values' field in the Pivot Table settings is currently set to 'COUNTA of Price'. A dropdown menu is open, showing various aggregation functions. The 'Average' function is highlighted. The dropdown menu includes the following options: SUM, COUNTA, COUNT, COUNTUNIQUE, AVERAGE, MAX, MIN, MEDIAN, PRODUCT, STDEV, STDEVP, VAR, and VARP. Below the dropdown, there is a 'Filters' section with an 'Add' button.

9. The pivot table is ready. You can also change the name of this table.



Vehicle_type	AVERAGE of Price
Car	26.319975
Passenger	27.7632
Grand Total	27.39075484

10. To create one more pivot table, go back to the data. Select the entire data, click on **Insert**, and then click on **Pivot table**.



type	Price_in_Thousands	Engine_size	Horsepower	Wheelbase	Width	Length	Curb_weight	Fuel_efficiency
er	21.5	1.8	140	101.2	67.3	172.4	2,639	
er	28.4	3.2	225	108.1	70.3	192.9	3,517	
er		3.2	225	106.9	70.6	192	3,47	
er	42	3.5	210	114.6	71.4	196.6	3,85	
er	23.99	1.8	150	102.6	68.2	178	2,998	
er	33.95	2.8	200	108.7	76.1	192	3,561	
er	62	4.2	310	113	74	198.2	3,902	
er	26.99	2.5	170	107.3	68.4	176	3,179	
er	33.4	2.8	193	107.3	68.5	176	3,197	
er	38.9	2.8	193	111.4	70.9	188	3,472	
er	21.975	3.1	175	109	72.7	194.6	3,568	
er	23.3	1.8	140	102.6	72.7	194.2	3,543	
er	31.985	3.8	205	113.8	74.7	206.8	3,778	
er	27.885	3.8	205	112.2	73.5	200	3,591	
er	39.895	4.6	275	115.3	74.5	207.2	3,978	
er	44.475	4.6	275	112.2	75	201		
er	39.685	4.6	275	108	75.5	200.6	3,843	

11. Click on **Create** to start creating the pivot table.

Car sales data

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1158 Manufacturer

Manufacturer	Model	Sales_in_Thoms	year	passenger	include
Acura	Integra	16,919	16.36	Passenger	
Acura	TL	39,384	19.875	Passenger	
Acura	CL	14,114	18.225	Passenger	
Acura	RL	8,588	29.725	Passenger	
Audi	A4	20,397	22.255	Passenger	
Audi	A6	18,78	23.555	Passenger	
Audi	A8	1,38	39	Passenger	
BMW	320i	19,747		Passenger	
BMW	520i	9,231	28.675	Passenger	
BMW	520i	17,527	36.125	Passenger	
Buick	Century	91,561	12.475	Passenger	
Buick	Regal	39,35	13.74	Passenger	
Buick	Park Avenue	27,851	20.19	Passenger	
Buick	LeSabre	83,257	13.36	Passenger	
Cadillac	DeVille	63,729	22.525	Passenger	
Cadillac	Seville	15,943	27.1	Passenger	
Cadillac	Eldorado	6,536	25.725	Passenger	

Create pivot table

Data range: Sheet2!\$1:\$58

Insert to:

☒ New sheet

☐ Existing sheet

Cancel Create

Manufacturer	Width	Length	Curb_weight	Fuel
Acura	181.2	67.3	172.4	2.635
Acura	108.1	70.3	192.9	3.517
Acura	106.9	70.6	192	3.47
Acura	114.6	71.4	196.6	3.85
Audi	102.6	68.2	178	2.998
Audi	108.7	76.1	192	3.561
Audi	113	74	198.2	3.902
BMW	107.3	68.4	176	3.179
BMW	107.3	68.5	176	3.187
BMW	111.4	70.9	188	3.472
Buick	109	72.7	194.6	3.368
Buick	109	72.7	196.2	3.543
Buick	113.8	74.7	206.8	3.778
Buick	112.2	73.5	200	3.591
Cadillac	115.3	74.5	207.2	3.978
Cadillac	112.2	75	201	
Cadillac	108	75.5	200.6	3.843